Banking Crisis Early Warning System

Project summary and validation audit

Bottom line: the pipeline runs end-to-end from the supplied processed panel, with leakage-safe country-grouped validation. The LSTM is the strongest evaluated model, but the small event count and incomplete raw World Bank coverage mean this is a research screening tool, not a deployment-ready policy engine.

Project Scope

The system predicts the first year of a systemic banking crisis across 24 emerging-market countries from 1971 to 2025. It contains 1,320 country-years and 46 crisis onsets (3.5% prevalence). Features are lagged one year, so the target-year crisis label is never paired with same-year macroeconomic information.

Component	Verified design
Target	First year of a systemic banking crisis episode
Predictors	Six lagged macro-financial indicators, including FX depreciation
Validation	Six crisis-balanced, country-grouped outer folds with inner validation
Models	Logistic regression, random forest, XGBoost, LightGBM, and LSTM

Data Readiness

The supplied model-ready panel passed shape, target-count, fold-assignment, and leakage tests. Feature missingness remains material: 10% for FX depreciation and roughly 36-55% for the World Bank indicators. The raw World Bank workbook covers only 16 of the 24 selected countries; the code now stops rather than silently rebuilding an incomplete panel. A complete 24-country World Bank re-export is required for fully raw-data reproducibility. A separate official-API enrichment extract now supplies coverage-audited banking and external-vulnerability candidates.

Validation and Modeling Audit

Controls added and verified in the final workflow

Control	Why it matters	Status
One-year feature lag	Prevents target-year information from entering an early-warning score	Verified
Country-grouped folds	Keeps all years of each country in exactly one outer partition	Verified
Nested validation	Keeps model selection and threshold selection away from outer test countries	Verified
Past-only filling	Uses country history, then train-only same-year and expanding fallbacks	Verified
Earliest-year gaps	Uses a zero-information placeholder only when no historical value exists	Verified
Raw coverage guard	Prevents incomplete source files from overwriting the model-ready panel	Added

Evaluation Interpretation

PR-AUC is the primary ranking metric because crisis onsets are rare. A random score would yield a PR-AUC near 0.035. AUC remains useful for broad rank ordering but can look optimistic in an imbalanced problem. Reported uncertainty is the standard deviation across the six held-out country folds, not a confidence interval.

Model Results

Model	AUC	PR-AUC	Assessment
LSTM	0.664 +/- 0.083	0.112 +/- 0.024	Best observed
LogReg	0.614 +/- 0.159	0.103 +/- 0.056	Below LSTM
RandomForest	0.585 +/- 0.069	0.091 +/- 0.056	Below LSTM
XGBoost	0.590 +/- 0.072	0.057 +/- 0.026	Below LSTM
Basel benchmark	0.547 +/- 0.077	0.051 +/- 0.015	Useful benchmark
LightGBM	0.571 +/- 0.081	0.048 +/- 0.011	Below LSTM

The LSTM achieved the highest held-out PR-AUC (0.112). Logistic regression was competitive (0.103), which is valuable evidence that simple, transparent relationships remain strong in this small sample. The fixed LightGBM reference configuration did not exceed the credit-to-GDP benchmark on PR-AUC and should not be selected for deployment without additional tuning and calibration.

Diagnostics and Recommendation

LightGBM Alert Threshold Diagnostics

Thresholds were chosen within each outer fold using inner-validation F2 optimization, then applied once to held-out countries. Mean held-out recall was 67.6% at precision of 4.1%, with a 57.3% alert rate. This is recall-oriented screening behavior, not a sufficiently precise autonomous alerting rule.

Metric	Average across six held-out folds
Recall	67.6%
Precision	4.1%
F2 score	0.164
Alert rate	57.3%

Reference LightGBM Drivers

Rank	Feature	Mean absolute SHAP value
1	inflation (t-1)	0.976
2	exchange rate depreciation	0.748
3	credit to gdp (t-1)	0.545
4	current account gdp (t-1)	0.325

External-Vulnerability Feature Screen

The new official World Bank extract supports two testable external indicators: debt service to exports (73.9% crisis-onset coverage) and reserves to short-term debt (69.6%). Adding both to LightGBM increased AUC to 0.606 and PR-AUC from 0.048 to 0.059. The PR-AUC gain is modest relative to variation across folds, so this remains an experimental extension rather than a replacement for the LSTM.

Recommended Next Steps

1. Re-export World Bank indicators for all 24 countries and rerun the complete raw-data pipeline. 2. Treat the LSTM and logistic regression as candidate research models, then add calibration, probability stability checks, and decision-cost analysis before choosing an operating threshold. 3. Add an external holdout period or new-country holdout before deployment claims. 4. Investigate twin-currency-crisis features only after they are available without leakage. 5. Maintain a human review queue for every model alert; current precision is too low for automated intervention.